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Dear Editorial Board,

Journal of Cryptography and Security Research

Re: Submission of manuscript — "V2 Inverted Hash 512×512: A Cryptographic Hash Function with Double-Scale Sacred Pair Inversion and Superlinear SAC Scaling"

I am writing to submit for your consideration the research article titled V2 Inverted Hash 512×512, the sixth work in an ongoing research program on pyramidal cryptographic hash functions initiated by the author. This design scales the 512×512 Inverted Hash architecture to a full 512-byte input, state, and output — doubling both the Sacred Pair K2 active region and the output space, while preserving the inverted tip-to-base anchor order that defines this branch of the research series.

The central finding of this work is the discovery of superlinear SAC scaling: when the architecture is scaled from 256-byte to 512-byte operation (doubling the K2 active anchors from 127 to 255), the Strict Avalanche Criterion deviation does not double as expected — it improves proportionally. The 256×256 Original Hash achieves SAC 0.98% over 2,048 bits; proportional expectation for 4,096 bits would be 1.96%; the V2 Inverted Hash achieves 1.12%, a 0.84% better result than the proportional prediction. This suggests that the K2 Sacred Pair inversion produces superlinear diffusion benefits as the architecture scales.

Eleven independent statistical tests on the complete 4096-bit output demonstrate: (1) avalanche effect of 49.894% with the tightest standard deviation in the research series — $\sigma = 0.741\%$; (2) Shannon entropy of 7.9987 bits/byte (99.98% efficiency); (3) Chi-square of 252.52, the lowest recorded in the series; (4) Pearson correlation of 0.02528 over 130,816 byte pairs at N=1,000 samples; (5) SAC deviation of 1.12% across 32,768 output bit positions; and (6) zero collisions in 400 attempts.

The article explicitly states the scope of this research: this is an experimental prototype for academic exploration of novel hash architectures. No claim of equivalence to established standards is made, and a formal cryptographic audit is strongly recommended before any production use.

I sincerely thank the editorial committee for their time and consideration, and remain available for any clarification or additional information required.

Sincerely,

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Project: V2 Inverted Hash 512×512 — Research Program in Pyramidal Cryptography

Manuscript enclosed: v2_inverted_hash_512x512.docx